DS3010 Midterm 2 (Practice)

Total: 100 Points

Name: In Class Solutions

Part 1: Single Choice Questions

1. In model selection, what is the primary goal?

   A. Minimize training error
   B. Maximize model complexity → only want to keep significant predictors
   C. Minimize expected test error
   D. Eliminate all predictors

2. Best subset selection is computationally expensive because:

   A. It requires cross-validation
   B. It evaluates all possible predictor combinations
   C. It uses regularization
   D. It fits only one model

3. What is the interpretation of a dummy variable coefficient? →

   A. The slope of Y vs X
   B. The difference relative to a reference group
   C. The variance of the category
   D. The predicted value of Y

$$Y \sim \beta_0 + \beta_1 \cdot X_1$$

$$X_1 = \begin{cases} 1 & \text{if college} \\ 0 & \text{no college} \end{cases}$$

Baseline: highschool

$$Y \sim \beta_0$$

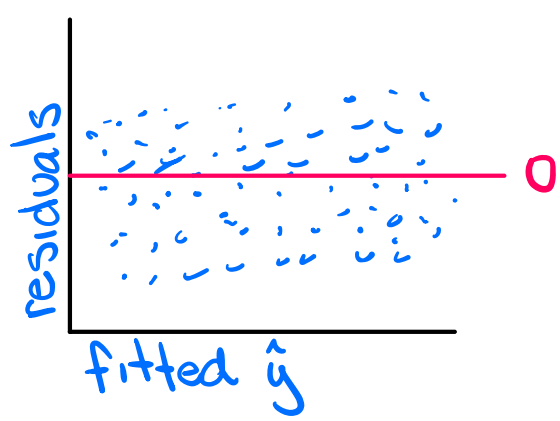
4. Which is a common indicator of multicollinearity?

   A. Low residuals
   B. High VIF values
   C. Small sample size
   D. Large intercept

Homoscedasticity: constant variance

5. Heteroscedasticity means:

   A. Errors are correlated
   B. Errors have non-constant variance
   C. Predictors are correlated
   D. Model is nonlinear



larger = heavier penalty

$$\left[RSS + \lambda \cdot \sum_{i=1}^{p} [\beta_i]^2\right] \leftarrow \text{minimize}$$

as close to zero as possible

$\beta_j \to 0$, change in y is less sensitive to change in x → lower var.

6. Increasing λ in regularization leads to:

   A. Lower bias, higher variance
   B. Higher bias, lower variance
   C. Lower bias, lower variance
   D. No change

7. Model selection

   Use the following output to carry out (1) subset selection, (2) forward selection, and (3) backward selection. The entire data set has 10,000 observations. For each algorithm, report the best model of each size. Choose one criterion, state and compute the criterion you will use to select your final model. Report the best model of each size in the table below. Report your final model for each algorithm.

$BIC = n \cdot \log\left[\frac{RSS}{n}\right] + p \cdot \log(n)$

Model	$\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$ (RSS)	BIC
A. $Y \sim X_1$	17694.34	5725.018
B. $Y \sim X_2$	17756.22	5729.929
C. $Y \sim X_3$	17987.45	5889.313
D. $Y \sim X_4$	17345.91	5526.137
E. $Y \sim X_1 + X_2$	17488.45	
F. $Y \sim X_1 + X_3$	17348.93	
G. $Y \sim X_1 + X_4$	17224.89	
H. $Y \sim X_2 + X_3$	17568.92	
I. $Y \sim X_2 + X_4$	17126.36	
J. $Y \sim X_3 + X_4$	17200.47	
K. $Y \sim X_1 + X_2 + X_3$	17248.56	
L. $Y \sim X_1 + X_2 + X_4$	17018.66	
M. $Y \sim X_1 + X_3 + X_4$	17007.83	
N. $Y \sim X_2 + X_3 + X_4$	17076.82	
O. $Y \sim X_1 + X_2 + X_3 + X_4$	17006.11	

	Best Subset		Forward		Backward	
Model size	**Model**	**Criteria result**	**Model**	**Criteria result**	**Model**	**Criteria result**
M_1 = 2	D	5526.137	D	5526.137	D	
M_2 = 3	I	5407	I	5407	J	
M_3 = 4	M	5347.73	L	5354.09	M	5354.09
M_4 = 5	O	5355.93	O	5355.93	O	5355.93
Final Model	M		L			

Part 3: Coding Problems

Problem 1 (Simulation Study: Ridge Bias)

Suppose the true population model is:

$$Y_i = 2 + 3X_{i1} + 5\log(X_{i2}) + \varepsilon_i, \quad \varepsilon_i \sim N(0,1)$$

You can generate the data using:

```
n = 100
X1 = seq(0,10,length.out = n)
X2 = runif(n)
error = rnorm(n,0,1)
Y = 2 + 3*X1 + 5*log(X2) + error
```

(a) Fit a ridge regression model with $\lambda = 2$.

(b) Repeat the simulation many times (e.g., 500 or 1000 times) and compute the average estimates of β_0, β_1, and β_2.

(c) Compute the bias for each parameter:

$$\text{Bias}(\hat{\beta}_j) = E[\hat{\beta}_j] - \beta_j$$

(d) Report your estimated biases:

Parameter	Bias
β_0	
β_1	
β_2	

(e) Briefly comment on your results. Does ridge introduce bias?

Problem 2 (Model Selection)

Now generate additional predictors:

```
X3 = rnorm(n)
X4 = rnorm(n)
```

(a) Apply subset selection (best subset, forward, or backward). Use AIC or BIC as your selection criterion.

(b) Report your selected model:

(c) Does your selected model recover the true model? Comment on your result.

Coding Questions

↳ check the assumptions

↳ subset selection

↳ ridge / lasso / bootstrap

Content: modules 3 - 5

question: one of the questions from lb on hw 8

question 7 will be in the exam as a coding question where the RSS values will be given as a vector